

# Mohit Kumar Sharma

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## Experience

### Software Engineer

Microsoft

Hyderabad, India

July 2024 – Present

- Architected **multi-tier caching layer** (in-memory + persistent on-disk with asynchronous background revalidation) and parallelized codec DLL probing using multithreaded concurrency patterns for a real-time communications media stack, reducing encoder/decoder capability-enumeration latency from **1-3s to 5ms** during application startup.
- Enhanced Teams H.264 encoder and decoder hardware acceleration through large-scale telemetry analysis, root cause debugging, experimentation, and production rollouts, delivering a **4% increase in decoder** hardware utilization, and **5% increase in encoder** hardware utilization, while lowering fallback rates and improving reliability.
- Engineered a scalable distributed test execution and analytics pipeline with real-time test tracking, automated failure analysis, intelligent alerting, and centralized dashboards, significantly **cutting manual debugging effort by 70%** and improving large-scale issue detection and overall test infrastructure reliability.
- Introduced telemetry metrics tracking hardware failures and encoder/decoder frame drops, and built **automated anomaly detection** on HW-utilization and reliability KPIs, enabling proactive regression detection and shrinking time-to-detect by over **90%**.
- Built an **AI-powered** investigation platform automating KPI regression analysis by correlating telemetry signals with diagnostic logs, orchestrating DDL experiments, and generating root-cause insights, decreasing manual investigation effort by 80% per incident and accelerating issue resolution.
- Technologies:** C++, Python, Multithreading & Concurrency, System Design, MCP, Agentic AI, KQL, Power BI.

### Software Engineer Intern

Microsoft

Hyderabad, India

May 2023 – July 2023

- Designed and developed multi-turn conversational interactions for the TAC Co-pilot, enhancing UI/UX by implementing intuitive chiclet-based input options and aligning the interface with the M365 Co-pilot design.
- Leveraged the **internal LLM framework** to accurately interpret user queries by **identifying intents** and **extracting entities**, improving overall conversational performance and user experience for the TAC Co-pilot.
- Technologies:** React, TypeScript, REST APIs, UI/UX, Prompt Engineering.

### Full-Stack Developer Intern

Warelogg

Remote

Dec 2022 – Dec 2022

- Enhanced the Warelogg React web application by improving UI/UX, responsiveness, and overall functionality, delivering a smoother end-user experience.
- Developed **5+ Node.js REST APIs** to manage warehouse inventory and administrative operations, implementing **role-based access control (RBAC)** to ensure secure data management and authorized access to business-critical resources.
- Technologies:** React, Node.js (Express.js), MongoDB, Tailwind CSS.

## Education

### Indian Institute of Technology (BHU) Varanasi

B.Tech - Electronics Engineering

2020 - 2024

CGPA - 8.42

## Projects

### Picture Manager Application [🔗](#)

April 2022 – May 2022

- Built a full-stack web application as part of the **Engage'22 program**, with **JWT-based authentication and authorization** and **facial-recognition login**, protecting all API endpoints and safeguarding user/admin access.
- Integrated intelligent photo-management functionality using **facial recognition** to automatically categorize images into individual and group folders, and implemented event-related CRUD operations with MongoDB for optimized data handling.
- Technologies:** Next.js, Node.js (Express.js), Face-api.js, MongoDB, Tailwind CSS.

## Technical Skills

- Languages & Frameworks:** C/C++, Python, JavaScript, React, NodeJS, SQL, NoSQL.
- Core Skills:** Full-Stack Development, DSA, OOPS, DBMS, System Design, CI/CD, Agile/Scrum, Problem-Solving.
- Tools & Platforms:** Visual Studio/Code, Postman, Git/Github, Claude, GitHub Copilot, Azure DevOps.

## Achievements

- Solved **800+ coding challenges** on platforms including LeetCode and GeeksforGeeks, demonstrating strong Data Structures & Algorithms problem-solving skills.